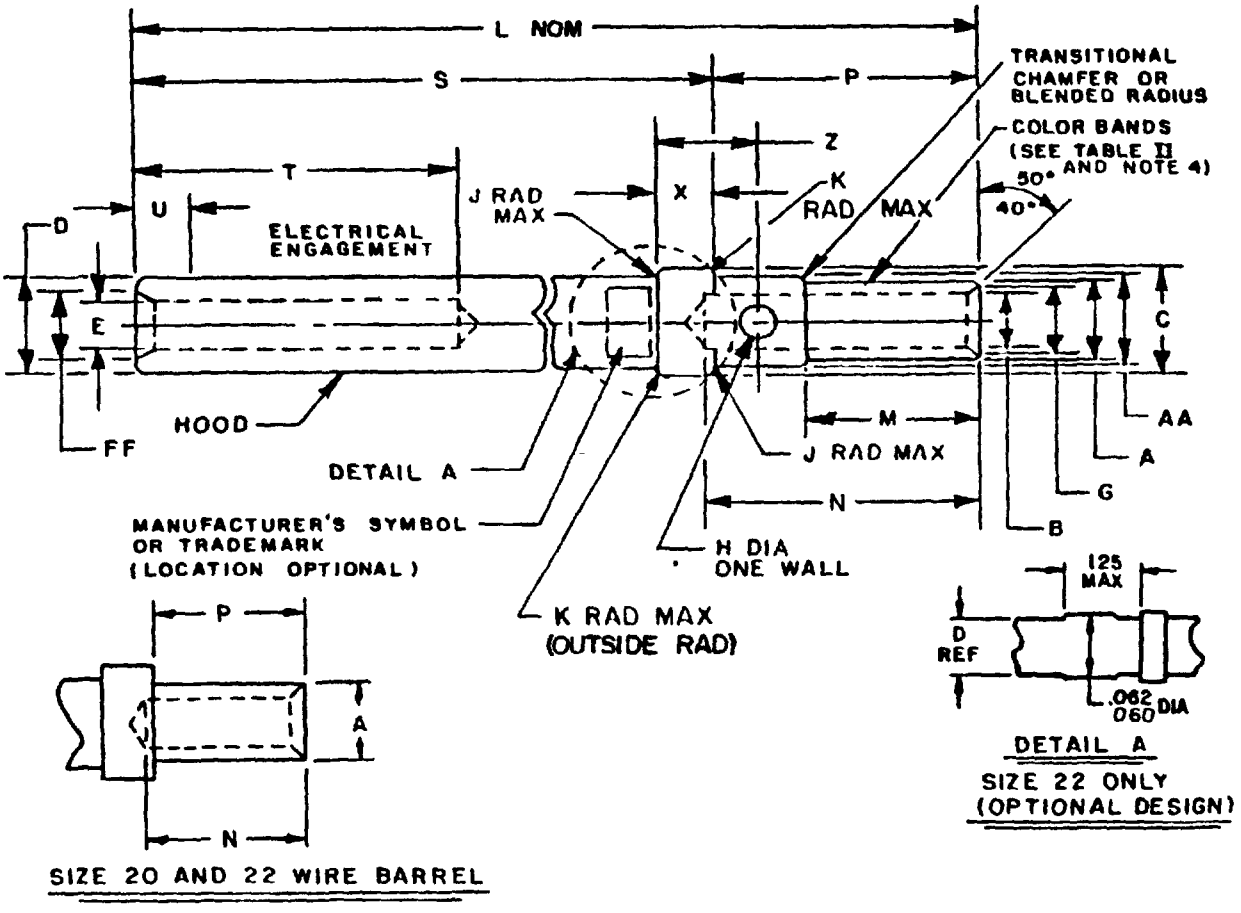
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE,
(FOR DOD-C-83527, CONNECTOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



Ⓐ FIGURE 1. Socket contact.

Ⓐ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:44Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.054	1.37	.083	2.11	.150	3.81
.010	0.25	.055	1.40	.084	2.13	.155	3.94
.018	0.46	.056	1.42	.086	2.18	.158	4.01
.025	0.64	.058	1.47	.090	2.29	.161	4.09
.029	0.74	.059	1.50	.0955	2.426	.165	4.19
.031	0.79	.060	1.52	.098	2.49	.186	4.72
.033	0.84	.061	1.55	.101	2.57	.190	4.83
.0340	0.864	.0615	1.562	.102	2.59	.231	5.87
.036	0.91	.0618	1.570	.103	2.62	.237	6.02
.0375	0.953	.062	1.57	.107	2.72	.245	6.22
.041	1.04	.0620	1.575	.110	2.79	.250	6.35
.0415	1.054	.0623	1.582	.111	2.82	.257	6.53
.042	1.07	.064	1.63	.113	2.87	.290	7.37
.043	1.09	.065	1.65	.116	2.95	.325	8.26
.044	1.12	.066	1.68	.130	3.30	.468	11.89
.0460	1.168	.068	1.73	.132	3.35	.630	16.00
.047	1.19	.069	1.75	.134	3.40	.650	16.51
.048	1.22	.070	1.78	.135	3.43	.918	23.32
.050	1.27	.080	2.03	.140	3.56	.965	24.51
.052	1.32	.082	2.08	.148	3.76		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. See MIL-C-39029 for option.

Ⓐ FIGURE 1. Socket contact - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:44Z
Check the source to verify that this is the current version before use.

Ⓐ TABLE I. Dimensions.

BIN code	A Dia	B Dia	C Dia	D Dia	E Dia (min only)	G Dia	H Dia	J Rad	K Rad	L (Nom)
601	.052	.0375	.070 (1.78)	.061	.031	.047	.022	.002	.003	.650 (16.51)
(22-22)	.050	.0340	.068 (1.73)	.059		.041	.018			.630 (16.00)
602	.068	.0460	.084 (2.13)	.074	.0415	.058	.033	.002	.003	.918 (23.32)
(20-20)	.065	.0430	.082 (2.08)	.068		.052	.025			.898 (22.81)
603	.103	.068	.134 (3.40)	.113	.064	.090	.042	.003	.005	.965 (24.51)
(16-16)	.101	.066	.130 (3.30)	.106		.080	.036			
604	.151	.102	.190 (4.83)	.161	.0955	.140	.042	.003	.005	.965 (24.51)
(12-12)	.148	.098	.186 (4.72)	.158		.130	.036			

BIN code	M Min	N	P	S	T Min	U Max	X	Z	AA Dia	FF Min
601	N/A	.165	.237	.410	.280	.043	.034	.161	---	.043
(22-22)		.135	.231	.401			.029	.132		
602	N/A	.190	.165	.750	.325	.041	.034	.069	N/A	.056
(20-20)		.150	.155	.745			.029	.054		
603	.150	.290	.257	.718	.468	.060	.048	.083	.111	.086
(16-16)		.250	.245	.710			.044	.069	.107	
604	.150	.290	.257	.718	.468	.060	.048	.083	.158	.116
(12-12)		.250	.245	.710			.044	.069	.155	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:44Z
Check the source to verify that this is the current version before use.

TABLE II. Design characteristics.

	BIN code	Color bands 1st	2nd	3rd	Mating end size	Wire barrel size	Type	Class
Ⓐ	601	Blue	Black	Brown	22	22	A	B
Ⓐ	602	Blue	Black	Red	20	20		
	603	Blue	Black	Orange	16	16		
	604	Blue	Black	Yellow	12	12		

TABLE III. Tools.

	BIN code	Basic crimping tool	Positioner	Installing tool	Removal tool
Ⓐ	601	M22520/2-01	M22520/2-23	M81969/1-01	M81969/1-01
Ⓐ	602	M22520/2-01 M22520/7-01	M22520/2-08 M22520/7-02	M81969/1-02	M81969/1-02
Ⓐ	603	M22520/1-01 M22520/7-01	M22520/1-02 Blue M22520/7-03	M81969/1-03	M81969/1-03
Ⓐ	604	M22520/1-01	M22520/1-11		M81969/28-02

REQUIREMENTS:

Dimensions, design characteristics, and configuration: See figure 1 and tables I and II.

Tools: See table III.

Vibration: Method 2005 of MIL-STD-1344, test condition IV.

Shock (specified pulse): Method 2004 of MIL-STD-1344, test condition A.

Ⓐ Mating contact: MIL-C-39029/93A.

Low signal level contact resistance: The mating depth shall be 0.180 inch.

Contact and engagement and separation forces: See table IV.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

International interest: NEPR 57.

Ⓐ Part number example: M39029/94-601.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:44Z
Check the source to verify that this is the current version before use.

Ⓐ TABLE IV. Contact engagement and separation forces.

BIN code	Mating end size	Initial			After conditioning		
		Minimum separation force (ounces)	Maximum average engagement force (ounces)	Maximum engagement force (ounces)	Minimum separation force (ounces)	Maximum average engagement force (ounces)	Maximum engagement force (ounces)
		Minimum diameter MS3197 pin	Maximum diameter MS3197 pin	Maximum diameter MS3197 pin	Minimum diameter MS3197 pin	Maximum diameter MS3197 pin	Maximum diameter MS3197 pin
601	22	0.75	2.50	3.0	.60	2.75	4.0
602	20	0.75	5.0	6.0	.60	6.00	7.0
603	16	2.00	9.0	12.0	1.0	11.5	15.0
604	12	3.00	12.0	15.0	2.0	14.5	18.0

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC, MC, SH
Air Force - 11, 99
DLA - ES

User activity:
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3648-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:44Z
Check the source to verify that this is the current version before use.